Open archaeology in Catalonia: challenges, barriers, and potential solutions

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Introduction

The benefits of making scientific data open both for the research community and society in general have been widely discussed. Archaeology, situated at the intersection of social sciences, natural sciences, and humanities, has been engaging with open data for several decades now (Hansen, 1992), and more and more institutions and funding agencies are advocating for this model.

As a highly collaborative discipline that studies cultural heritage, receives mostly public funding for research, and generally uses a destructive methodology such as excavation, archaeology seems perfectly suited for the open data model. This model has been considered beneficial, even necessary, for the advancement of archaeological knowledge (Kintigh, et al., 2014) and is seen for some as a moral imperative for the discipline (Marwick et al., 2017). The general belief is that open data will bring increased sustainability and improve archaeological research (Atici et al., 2013; Beck & Neylon, 2012; Kansa et al., 2014; Moore & Richards, 2015).

However, this model has also faced criticism regarding its implications for the discipline and its knowledge-making process (Huggett, 2014; 2015; 2023), especially because of the digitalization and datafication that these practices inevitably entail (Haciguzeller et al., 2021). These ramifications are particularly significant from a data management perspective, as the growing digitalization of the archaeological record poses a challenge for its curation and long-term preservation. At the same time, it also opens the door for higher findability, accessibility, and sharing options, that make reuse of archaeological data possible.

Despite significant efforts to build infrastructure to support data publication over the last decade (Wright & Richards, 2018), the adoption of open data remains limited in the discipline (Karoune, 2022). In face of this situation, research has recently revolved around technical issues such as infrastructure or data description, often at an institutional level (Jakobsson et al., 2021; 2023). This focus, while indispensable for promoting the implementation of the model, has long disregarded researchers' habits and perspectives, failing to understand the social variables that intersect with research practices.

Goal

The goal of this long paper is to identify the challenges that the implementation of open data is facing in archaeology, focusing on a specific, well-defined research community: the Catalan archaeological research community. Following Levin and Leonelli's (2017) idea that the open model cannot be a one-size-fits-all model and must be adapted to the singularities of each research community of practice or system of research practice (Leonelli, 2022), we aim to understand the research community in depth, including its traditions, its actors, their practices, and all the supporting infrastructure. This involves studying two main aspects of the research ecosystem: infrastructure —comprising physical infrastructures (data portals, platforms, and repositories) as well as human (the community of research and the institutions behind it) and social (the laws and norms regulating archaeological research and data)—, and researchers' practices and perspectives.

By approaching the social and physical context of archaeological research, we seek to identify the factors preventing adherence to open practices and find a path for a smooth, successful transition towards more open research. The final aim of this paper is to draw up a set of recommendations or future steps to promote the model in this specific context.

Methodology

The research presented in this long paper uses two main methodologies, one for each of the two objects of study: archaeological research infrastructure and researchers' practices and perspectives.

On one hand, we analyse the current infrastructures that support archaeological data in Catalonia, including the laws and norms that regulate archaeological research, the main institutions, the principal funding bodies, and the platforms and physical infrastructure that support the management and publication of research data. All these infrastructures have been described and evaluated through a custom set of indicators, with the

goal to provide a standardized, objective, and comparable framework.

On the other hand, we address the practices and perspectives around data management and sharing among Catalan archaeologists. The approach selected has been qualitative research, due to the small size of the population and the will to better attain the researchers reasoning and opinions behind their actions. To do so, 14 in-depth semi-structured interviews have been conducted with principal investigators of archaeological research projects. Their responses have been then transcribed verbatim and analysed with qualitative content analysis (Schreier, 2012) using a mixed coding frame build both inductively from the literature (Zuiderwijk et al., 2020), and deductively from the content itself.

Results

The research allows us to delineate the current situation of archaeological data in this region and to identify the needs of researchers, as well as the current barriers inhibiting archaeological data openness. The results reveal an active data-sharing tradition based on bilateral sharing and a strong research culture that naturally repels open practices, despite most researchers personally aligning with the principles of openness behind these new models. These traditions, rooted in the lack of trust of researchers, sustained by the current infrastructure and the community organization, and fed by the discipline's unique characteristics, seem to perpetuate themselves.

The study highlights the dual nature of archaeological research, situated between science and heritage laws and infrastructures. This duality poses a problem when managing and sharing archaeological data, as basic concepts such as protection, ownership, and preservation responsibility appear blurry and undefined. In this sense, the lack of clarity in current legislation and norms, as well as preservation concerns, acts as a barrier for open data. Among the other factors perceived as inhibitors, key findings include the importance of excavation reports, the identification of social and personal barriers to open data adoption, and a general lack of awareness on data management and open data best practices.

From these results, a set of recommendations addressed to the main actors of Catalan archaeological research have been developed which seek to meet the particular needs of the community and the current situation or archaeological research in Catalonia. This should allow to foster and encourage the adoption of the open data model and ensure an easy transition towards more sustainable, open, and transparent archaeological research.

Conclusions

The transition to open data in archaeology is not only a technical challenge but also a social one. The findings of this study underscore the importance of considering the social dynamics and cultural practices within the research community, as the resistance to open data practice is related to long-standing traditions, not easily overcome by technical solutions alone.

One of the critical insights from this research is the need for an adapted approach to promoting open data. It is crucial to understand the research system's singularities to promote open data not only focusing on building infrastructure but also fostering a culture of openness and trust within the community.

Moreover, the study highlights the importance of involving researchers in the development and implementation of open data policies. By engaging with the research community and understanding their needs and concerns, more effective solutions can be developed.

The implications of these findings, while tailored to the specific context of Catalonia, extend beyond this region, offering valuable insights for other research communities. The recommendations developed in this paper provide a roadmap for promoting open data practices in Catalan archaeological research. However, they offer valuable insights that can be applied to other communities and, after deep investigation on the contextual characteristics of these other research systems, they could be adapted or serve as inspiration to promote open archaeological data.

In conclusion, by fostering a culture of openness and trust, building the necessary infrastructure, and involving researchers in the development of open data policies, we can ensure a successful transition to open data in archaeology.

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